


**Figure 5.2 A slot machine with an arm that can be pulled. The machine may return coins according to its reward probability when its arm is pulled.**

With this row of slot machines, a strategic player may turn this setting into a decision-making challenge and aim to try out the slot machines in some intelligent manner, so that she can most quickly identify the machine that has the highest reward probability. Her intent here is to maximize the amount of reward she receives, given that she can only pull the arms of these machines for a specific number of times.

**IMPORTANT** **Definition of multi-armed bandit**

**This decision-making problem is called multi-armed bandit (or MAB), as there are multiple arms that we may pull. The goal is to design a policy of deciding which machine whose arm we should pull next in order to maximize the total reward that we obtain at the end.**

We see that MAB possesses many characteristics of an optimization-making-under-uncertainty problem like BayesOpt:

- We can take specific actions: Each action corresponds to pulling the arm of a specific slot machine.
- We have a limited budget: We can only pull these arms for a specific number of times until we have to stop.
- There's uncertainty in the outcomes of the actions that may be taken: We don't know what the reward probability of each slot machine is, and we can't even estimate it until we have pulled its arm multiple times. Further, upon pulling an arm, we can't know for sure whether we will receive coins, as there's randomness in the reward.
- We'd like to optimize for an objective: We aim to maximize the *cumulative reward*, which is the total number of coins we receive while pulling the arms of these machines until we stop.

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